Intelligent Agent

Assignment: Designing and Evaluating an Intelligent Agent

Chosen Agent: Intelligent Facility Maintenance & Repair Prioritization Agent

This assignment develops and evaluates Maintain AI, an intelligent agent for facility maintenance. The design covers problem identification, text-based problem formulation, a perception-action model, PEAS analysis, and classification of the agent's operating environment.

1. Identify the Problem

Facility maintenance teams often receive several faults at the same time. In many organizations, requests are handled by report time or manual judgement, which can allow a low-impact issue to be addressed before a safety-critical or operationally important failure.

- **Reactive repairs** - equipment is often serviced only after performance has dropped or a breakdown has occurred.
- **Poor prioritization** - urgent faults can wait while less important requests consume available technicians.
- **Resource mismatch** - the right technician or spare part may not be available when the work order is created.

For example, if an elevator reports abnormal vibration, a classroom air conditioner stops cooling, and one corridor light fails at the same time, the system should not simply follow the reporting order. The elevator may require immediate escalation because of potential safety and service risk.

Problem Statement

A facility needs an intelligent agent that can combine maintenance complaints, equipment condition, historical failure data, operational impact, technician availability, and spare-part status to decide what should be repaired first, when a problem must be escalated, and when preventive maintenance should be scheduled.

2. Formulate the Problem (Text-Based)

The maintenance problem can be expressed through the agent's initial state, available actions, desired goal state, percepts, and the decision sequence it repeatedly follows.

Initial State

At the beginning of a decision cycle, the agent has a changing set of open work orders and facility assets. The true condition of each asset is only partially known until complaint details, sensor readings, service history, and current resource information are collected.

Actions Available to the Agent

- Rank or re-rank maintenance requests according to risk and operational impact
- Assign an appropriate technician or maintenance team
- Schedule corrective or preventive maintenance
- Escalate critical faults and issue alerts
- Continue monitoring or defer a low-risk task when justified

Goal State

High-risk and high-impact faults are addressed quickly, equipment downtime is minimized, preventable failures are detected early, and maintenance resources are used efficiently without ignoring lower-priority work.

Percepts (Inputs the Agent Receives)

- Maintenance complaint details: issue type, location, reported severity, and time of report
- Equipment condition: alarms, temperature, vibration, runtime, or other sensor readings
- Asset profile: age, criticality, previous breakdowns, repair history, and service interval
- Operational impact: number of people affected, building zone, occupancy, and operating hours
- Resource status: technician skills and availability, spare parts, and estimated repair time

Sequence of Actions (Path)

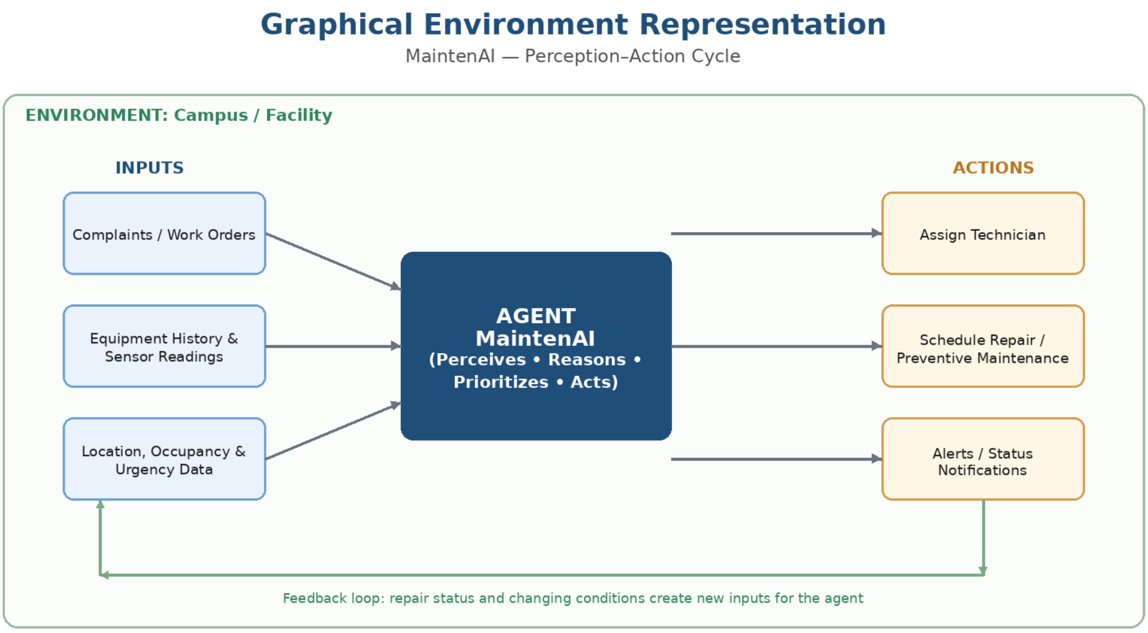
Receive complaint / sensor data -> identify the affected asset and location -> assess safety and operational risk -> calculate priority -> check technician and spare-part availability -> assign, schedule, or escalate -> complete repair/service -> collect feedback -> update asset history and priorities -> repeat

Formulated Problem Statement

Given active maintenance requests, equipment history, real-time condition information, operational impact, and current maintenance resources, the agent must decide which task should be handled first, which resource should be assigned, and whether preventive action or escalation is required, while continuously updating those decisions as the facility changes.

2.1 Graphical Representation of the Environment

The diagram below shows the perception-action cycle of Maintain AI. Inputs from maintenance systems, equipment sensors, asset records, occupancy information, and resource status are analyzed by the agent. The resulting priorities, assignments, schedules, alerts, and notifications act back on the facility, producing new information for the next decision cycle.



3. Evaluate in the Context of PEAS

PEAS describes the task environment of an intelligent agent using four elements: Performance measure, Environment, Actuators, and Sensors.

Component	Details
P - Performance Measure	Minimizes safety risk and downtime; speeds response to critical faults; improves uptime and preventive-maintenance compliance; and uses technicians and spare parts efficiently.
E - Environment	Campus or building facilities, occupied areas, HVAC, elevators, electrical and plumbing systems, maintenance staff, spare-parts inventory, and operating schedules.
A - Actuators	Re-prioritize work orders, assign technicians, create repair or preventive schedules, escalate critical incidents, issue alerts, and update work-order status.
S - Sensors	Maintenance tickets, IoT/BMS readings, equipment alarms, asset and repair history, occupancy data, technician availability, spare-part status, and repair feedback.

4. Identify the Environmental Properties

Maintain AI operates in an environment that changes continuously and is never fully known. Its standard environmental classifications are shown below.

Property	Type	Justification
Observability	Partially Observable	Complaints, sensors, and records do not reveal every hidden defect or future failure. Information can also be delayed, incomplete, or inaccurate.
Determinism	Stochastic / Uncertain	Results are not fully predictable: faults can worsen, parts can be delayed, technicians can become unavailable, and new readings can change priority.
Episodes	Sequential	Current choices affect later risk and resource availability. A delayed repair can become more serious, while assigning one technician changes what can be handled next.
Change	Dynamic	New faults, occupancy, sensor readings, equipment condition, and technician availability can change while the agent is deciding.
Values	Mixed (Discrete + Continuous)	Priority levels and work-order states are discrete, while temperature, vibration, runtime, energy use, and risk estimates can be continuous.
Agents	Single Agent	Maintain AI is the only autonomous reasoning agent in this model. Technicians, users, and managers are treated as external actors and resources.

End of Assignment – Maintain AI by Alyan